6 (a) The lowest electron energy levels in an isolated hydrogen atom are shown in Fig.6.1.

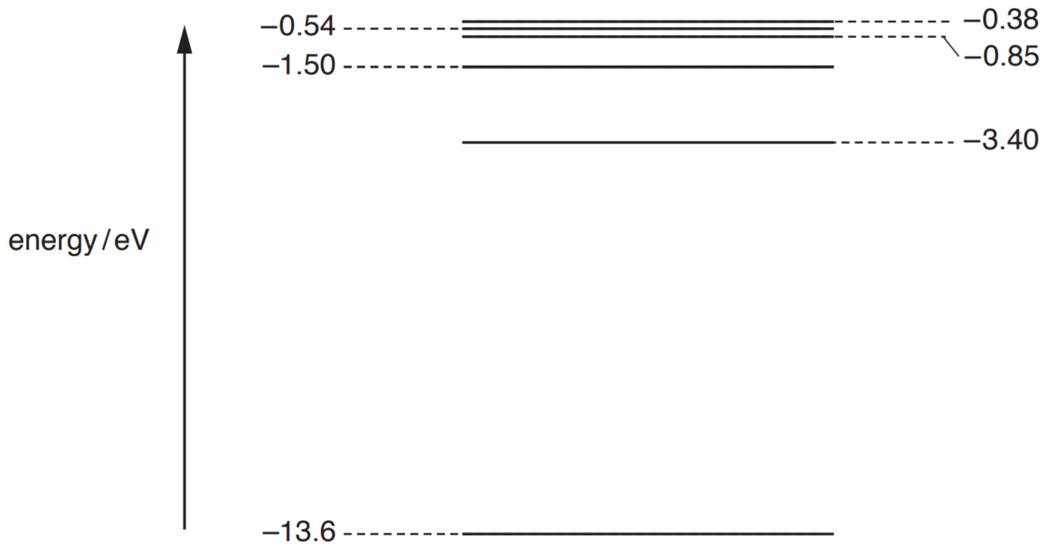

Fig. 6.1 (not to scale)

(i) An electron is initially at the energy level –0.85 eV. State the total number of different wavelengths that may be emitted as the electron de-excites (loses energy).

number = ……….……………… [1]

(ii) Photons resulting from electron de-excitation from the –0.85 eV energy level are incident on the surface of a sample of platinum.

Platinum has a work function energy of 5.6 eV.

Determine

1. the maximum kinetic energy, in eV, of a photoelectron emitted from the surface of the platinum,

maximum kinetic energy = ……………………… eV [2]

2. the wavelength of the photon producing the photoelectron in **(ii)1.**

wavelength = ……………………… m [2]

(b) Beta decay provides evidence for the existence of neutrinos. One such example is the spontaneous decay of carbon-14 into nitrogen-14. Neutrinos, ν, which have negligible mass and no charge, are emitted in the process.

$$^{14}_{6}C \rightarrow ^{14}_{7}N + e + \nu$$

In beta decays, the electrons emitted has a range of possible velocity. Explain how this can be used to predict the existence of neutrinos.

....

....

....

....

....

.... [3]

[Total: 8]